Melvyn Huynh

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WORK EXPERIENCE

Student Researcher **Sep 2026 – Present**
Human-centered Robotics & AI Group @Idiap Research Institute, part-time *Martigny, Switzerland*
- Build a **vision-based perception module** for Pepper that infers user engagement for responsive robot interaction
- Implement a **ROS4HRI pipeline** converting vision data into interpretable user-state primitives

Research Assistant **Sep 2026 – Present**
Applied AI Lab @UNIL, part-time *Lausanne, Switzerland*
- Build a **Reddit dataset** on how people running a business alone experience AI, collected with **LLM agents**

Embedded Machine Learning Engineer **Feb 2026 – Present**
N-Pulse, part-time *Lausanne, Switzerland*
- **Computer vision** (Sep 2026 – Present): Deploy **YOLO** vision model on Arduino UNO Q for on-device inference
- **Biosignal decoding** (Feb – Jul 2026): Integrated ML models for **EMG and EEG decoding** into embedded C++ firmware; benchmarked **TinyML** techniques for resource-limited hardware

AI Intern **Jul 2026 – Aug 2026**
Sparx (Virtuos Studio)* *Ho Chi Minh City, Vietnam*
- Built and deployed automated **invoice extraction** website via **vision LLM API**, replacing manual entry
- Implemented automated web scraping of relevant news delivered to team on a weekly basis

Robotics & ML Engineer **Mar 2026 – Jul 2026**
EPFL AI Team, part-time *Lausanne, Switzerland*
- Implement **imitation learning and RL pipelines** (LeRobot, PyTorch) on robotic arm leader-follower setup (SO-101)
- Collect training datasets via camera-based teleoperation and train policies for **robotic manipulation** tasks

EDUCATION

EPFL **2025 – 2028**
M.S., Robotics *Lausanne, Switzerland*
- **Relevant Coursework:** Deep Learning, Applied Machine Learning, Computer Vision, Deep Learning for Autonomous Vehicles, Aerial Robotics

EPFL **2021 – 2025**
B.S., Microengineering *Lausanne, Switzerland*
- **Relevant Coursework:** Embedded Systems, Signal Processing, Robotics, Project-Oriented Programming (C++)

TEACHING/MENTORING

Teaching Assistant **Sep 2026 – Present**
AI for Business, Applied AI Lab @UNIL, part-time
- Accompany students through labs and quizzes for the course taught by Prof. Yash Raj Shrestha

SKILLS

- **AI/ML:** PyTorch, TensorFlow, scikit-learn, imitation learning, deep learning, LLM API integration, ML pipelines
- **Computer Vision:** Object detection, image processing, sensor fusion, OpenCV
- **Programming:** Python, C++, C, MATLAB
- **Embedded Systems:** TinyML, firmware deployment (C++), real-time embedded boards, sensor data acquisition
- **Tools & Platforms:** Git, Linux, ROS2, Docker, sim-to-real transfer (Webots, MuJoCo), data analysis (Pandas, NumPy)
- **Language:** French (Native), Cantonese (Native), English (Bilingual, TOEIC C1), Vietnamese (B1), Mandarin (A2)

SELECTED COURSE PROJECTS

Deep Learning for Autonomous Vehicles
Feb – Jun 2026, EPFL
- Designed a **transformer-based trajectory prediction model** with auxiliary segmentation, ADE 1.52 on Kaggle
- Closed the **sim-to-real gap** with mixed synthetic/real training and domain-randomized augmentation, ADE 1.59

Autonomous Drone Navigation
Mar – May 2026, Aerial Robotics, EPFL
- Deploy autonomous flight algorithms on **Crazyflie drones** with **sim-to-real transfer** using Webots simulation
- Develop real-time navigation pipelines with **OpenCV-based gate detection** and flight control

CERTIFICATIONS & INTERESTS

Google IT Automation with Python (Google)
Interests: Piano (content creation), Football